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ECON1050 Tutorial 8 *Week 10*

Constrained Optimization

Tutorial 8 - Semester 1 2026

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Today's Plan

- ▶ Recap: key ideas from Lecture 8
- ▶ Tutorial questions: Q1–Q4

Recap: Equality-constrained optimization

General form

There are two parts to the problem: an **objective function** and an **equality constraint**.

$$\max \text{ or } \min f(x, y) \leftarrow \text{subject to } \boxed{g(x, y) \equiv c.}$$

Example: firm goal: max profit

In ECON 2050

Key idea

2, 3 constraint

- ▶ The constraint must be satisfied at the solution.
- ▶ The variables are mutually dependent along the constraint.
- ▶ This lecture focuses on one equality constraint.

Recap: The Lagrangian

Lagrange function

For

$$\max \text{ or } \min f(x, y) \quad \text{s.t.} \quad g(x, y) = c,$$

form the Lagrangian

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c).$$

Lagrange multiplier

- ▶ λ is the **Lagrange multiplier**.
- ▶ It is treated as an unknown constant.
- ▶ When the constraint is satisfied, $g(x, y) - c = 0$.

Recap: First-order conditions

FOCs

Differentiate the Lagrangian and set the partial derivatives equal to zero:

$$\left. \begin{aligned} \mathcal{L}_x &= f_x - \lambda g_x = 0, \\ \mathcal{L}_y &= f_y - \lambda g_y = 0, \\ \mathcal{L}_\lambda &= -(g(x, y) - c) = 0. \end{aligned} \right\} \Rightarrow \text{stationary point } (x^*, y^*, \lambda^*)$$

Recap: Interpreting λ

Shadow price

λ tells us how much the optimal value of the objective function changes when the right-hand side of the constraint increases by one unit.

Example: firm objective: profit
constraint: time or money

Optimal value function

Let

$$f^*(c) = f(x^*(c), y^*(c)),$$

where $x^*(c)$ and $y^*(c)$ solve the constrained optimization problem for a given c . Under regularity conditions,

$$\frac{df^*(c)}{dc} = \lambda(c).$$

Recap: Local second-order conditions

Definition

For a stationary point (x_0, y_0, λ) , define

$$D(x_0, y_0, \lambda) = \mathcal{L}_{11}(g_2)^2 - 2\mathcal{L}_{12}g_1g_2 + \mathcal{L}_{22}(g_1)^2,$$

where g_i is evaluated at (x_0, y_0) .

$\mathcal{L}_{11}, \mathcal{L}_{12}, \mathcal{L}_{22}$ are the 2nd partial derivatives
 g_1, g_2 1st pair of $g(x, y)$

Classification

$$D(x_0, y_0, \lambda) < 0 \Rightarrow \text{local maximum}, \quad D(x_0, y_0, \lambda) > 0 \Rightarrow \text{local minimum}.$$

Recap: Concave and convex Lagrangians

Sufficient conditions on a convex set

For a C^2 function $f(x, y)$ defined on a convex set S :

- ▶ If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is concave.
- ▶ If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is convex.

Global sufficient conditions

- ▶ If the Lagrangian is concave, a stationary point solves the maximization problem, which is a global max.
- ▶ If the Lagrangian is convex, a stationary point solves the minimization problem, which is a global min.

Tutorial Q1

Definition

For a stationary point (x_0, y_0, λ) , define

$$D(x_0, y_0, \lambda) = \mathcal{L}_{11}(g_2)^2 - 2\mathcal{L}_{12}g_1g_2 + \mathcal{L}_{22}(g_1)^2,$$

where g_i is evaluated at (x_0, y_0) .

Question 1

Solve the constrained optimization problem:

$$\max_{x_1, x_2} y = 2x_1x_2 \quad \text{s.t.} \quad 2x_1 + x_2 = 10.$$

Use the second-order condition to confirm the classification.

$$f(x_1, x_2) = 2x_1x_2, \quad g(x_1, x_2) = 2x_1 + x_2$$

$$L = 2x_1x_2 - \lambda(2x_1 + x_2 - 10)$$

$$\frac{\partial L}{\partial x_1} = 2x_2 - 2\lambda = 0 \Leftrightarrow x_2 = \lambda \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 2x_1 - \lambda = 0 \Leftrightarrow x_1 = \frac{\lambda}{2} \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = -(2x_1 + x_2 - 10) = 0 \Leftrightarrow 2x_1 + x_2 = 10 \quad (3)$$

plug (1) and (2) into (3):

$$2 \cdot \frac{\lambda}{2} + \lambda = 10 \Rightarrow \lambda^* = 5 \quad (4)$$

plug (4) into (1) and (2):

$$(1): x_2^* = \lambda^* = 5$$

$$(2): x_1^* = \frac{\lambda^*}{2} = \frac{5}{2}$$

stationary point $(x_1^*, x_2^*, \lambda^*) = (\frac{5}{2}, 5, 5)$

$$L_{11} = 0, \quad L_{12} = 2, \quad L_{21} = 2, \quad L_{22} = 0, \quad g_1 = \frac{\partial g}{\partial x_1} = 2, \quad g_2 = 1$$

$$D = 0 \cdot 1^2 - 2 \cdot 2 \cdot 2 \cdot 1 + 0 = -8 < 0$$

$\Rightarrow (\frac{5}{2}, 5)$ is a local minimum

Tutorial Q1: FOCs and stationary point

Knowledge used

- ▶ Form the Lagrangian $\mathcal{L} = f - \lambda(g - c)$.
- ▶ Use the FOCs $\mathcal{L}_{x_1} = \mathcal{L}_{x_2} = \mathcal{L}_\lambda = 0$.

Solution

$$\mathcal{L} = 2x_1x_2 - \lambda(2x_1 + x_2 - 10).$$

The FOCs are

$$2x_2 - 2\lambda = 0, \quad 2x_1 - \lambda = 0, \quad 2x_1 + x_2 - 10 = 0.$$

From $\lambda = 2x_1$ and $2x_2 - 2\lambda = 0$,

$$x_2 = 2x_1.$$

Substitute into the constraint:

$$2x_1 + 2x_1 = 10 \Rightarrow x_1^* = \frac{5}{2}, \quad x_2^* = 5, \quad \lambda = 5.$$

Tutorial Q1: Second-order condition

Knowledge used

- ▶ Use $D = \mathcal{L}_{11}(g_2)^2 - 2\mathcal{L}_{12}g_1g_2 + \mathcal{L}_{22}(g_1)^2$.
- ▶ $D < 0$ indicates a local maximum.

Solution

$$\mathcal{L}_{11} = 0, \quad \mathcal{L}_{22} = 0, \quad \mathcal{L}_{12} = 2, \quad g_1 = 2, \quad g_2 = 1.$$

Hence

$$D = (0)(1)^2 - 2(2)(2)(1) + (0)(2)^2 = -8 < 0.$$

Therefore,

$$(x_1^*, x_2^*) = \left(\frac{5}{2}, 5\right)$$

is a local maximum.

Tutorial Q2

Question 2

Solve the constrained minimization problem:

$$\min_{x_1, x_2} y = x_1^2 + 2x_2^2 - 2x_1x_2$$

$f(x_1, x_2)$

$$\text{s.t.} \quad 2x_1 - 7x_2 + 29 = 0.$$

$g(x_1, x_2)$

Then show that the solution is a global minimum.

$$L = x_1^2 + 2x_2^2 - 2x_1x_2 - \lambda(2x_1 - 7x_2 + 29)$$

$$\frac{\partial L}{\partial x_1} = 2x_1 - 2x_2 - 2\lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial x_2} = 4x_2 - 2x_1 + 7\lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = -(2x_1 - 7x_2 + 29) = 0 \quad (3)$$

$$0 \times \frac{7}{2}; 7x_1 - 7x_2 - 7\lambda = 0 \quad (4)$$

$$(4) + (2): 5x_1 - 3x_2 = 0$$
$$x_1 = \frac{3}{5}x_2 \quad (5)$$

plug (5) into (3):

$$\frac{6}{5}x_2 - 7x_2 + 29 = 0$$

$$\frac{6}{5}x_2 - \frac{35}{5}x_2 + 29 = 0$$

$$x_2^* = 5$$

$$\text{From (5): } x_1^* = \frac{3}{5}x_2^* = 3$$

$$\text{From (1): } \lambda^* = x_1 - x_2 = -2$$

Stationary point is $(3, 5, -2)$

For a C^2 function $f(x, y)$ defined on a convex set S :

► If $f_{xx} \leq 0$, $f_{yy} \leq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is concave.

► If $f_{xx} \geq 0$, $f_{yy} \geq 0$, and $f_{xx}f_{yy} - f_{xy}^2 \geq 0$ throughout S , then f is convex.

Global sufficient conditions

► If the Lagrangian is concave, a stationary point solves the maximization problem, which is a global max.

► If the Lagrangian is convex, a stationary point solves the minimization problem, which is a global min.

$$L_{11} = 2, L_{12} = -2, L_{21} = -2, L_{22} = 4$$

$$D = L_{11} \cdot L_{22} - L_{12}^2 = 8 - 4 = 4 > 0$$

$\Rightarrow L$ is convex

$\Rightarrow (3, 5, -2)$ is a global minimum

Tutorial Q2: FOCs and stationary point

Knowledge used

- ▶ Apply the Lagrange multiplier method to a minimization problem.
- ▶ Solve the three FOCs simultaneously.

Solution

$$\mathcal{L} = x_1^2 + 2x_2^2 - 2x_1x_2 - \lambda(2x_1 - 7x_2 + 29).$$

$$2x_1 - 2x_2 - 2\lambda = 0, \quad 4x_2 - 2x_1 + 7\lambda = 0, \quad 2x_1 - 7x_2 + 29 = 0.$$

From the constraint,

$$x_1 = \frac{7x_2 - 29}{2}.$$

Substitute into the first FOC:

$$5x_2 - 29 - 2\lambda = 0 \Rightarrow \lambda = \frac{1}{2}(5x_2 - 29).$$

Substitute into the second FOC:

$$-3x_2 + 29 + 7\lambda = 0 \Rightarrow x_2 = 5.$$

Then

$$x_1^* = 3, \quad x_2^* = 5, \quad \lambda = -2.$$

Tutorial Q2: Local and global minimum

Knowledge used

- ▶ Use the local second-order condition to classify the stationary point.
- ▶ Use convexity of the Lagrangian to show a global minimum.

Solution

$$\mathcal{L}_{11} = 2, \quad \mathcal{L}_{22} = 4, \quad \mathcal{L}_{12} = -2, \quad g_1 = 2, \quad g_2 = -7.$$

$$D = 2(-7)^2 - 2(-2)(2)(-7) + 4(2)^2 = 58 > 0,$$

so $(3, 5)$ is a local minimum.

For global minimization,

$$\mathcal{L}_{11} = 2 \geq 0, \quad \mathcal{L}_{22} = 4 \geq 0, \quad \mathcal{L}_{11}\mathcal{L}_{22} - (\mathcal{L}_{12})^2 = (2)(4) - (-2)^2 = 4 \geq 0.$$

The Lagrangian is convex, so

$$(x_1^*, x_2^*) = (3, 5) \text{ is a global minimum.}$$

Question 3

A firm produces two goods. It wants to maximize profit subject to a time constraint:

$$\max_{q_1, q_2} \pi(q_1, q_2) = TR - TC, \quad 2q_1 + 4q_2 = 100, \quad 7 \text{ min}$$

where

$$TR = 15q_1 + 12q_2, \quad TC = 20 - 4q_1q_2 + 7q_1 + 6q_2.$$

Find the profit-maximizing outputs, confirm the local maximum, and interpret λ .

Question 3

A firm produces two goods. It wants to maximize profit subject to a time constraint:

$$\max_{q_1, q_2} \pi(q_1, q_2) = TR - TC, \quad 2q_1 + 4q_2 = 100, \quad \leftarrow 7 \text{ min}$$

q_1, q_2

where

$$TR = 15q_1 + 12q_2, \quad TC = 20 - 4q_1q_2 + 7q_1 + 6q_2.$$

Find the profit-maximizing outputs, confirm the local maximum, and interpret λ .

$$\pi(q_1, q_2) = TR - TC$$

$$= 15q_1 + 12q_2 - (20 - 4q_1q_2 + 7q_1 + 6q_2)$$

$$= 8q_1 + 6q_2 + 4q_1q_2 - 20$$

$$g(q_1, q_2) = 2q_1 + 4q_2$$

$$L = 8q_1 + 6q_2 + 4q_1q_2 - 20 - \lambda(2q_1 + 4q_2 - 100)$$

$$\frac{\partial L}{\partial q_1} = 8 + 4q_2 - 2\lambda = 0 \quad (1)$$

$$\frac{\partial L}{\partial q_2} = 6 + 4q_1 - 4\lambda = 0 \quad \Leftrightarrow 3 + 2q_1 - 2\lambda = 0 \quad (2)$$

$$\frac{\partial L}{\partial \lambda} = -(2q_1 + 4q_2 - 100) = 0 \quad (3)$$

$$(2) - (1): 2q_1 - 4q_2 - 5 = 0 \quad (4)$$

$$(3) \times 2 + (4): 4q_1 - 105 = 0$$

$$q_1^* = \frac{105}{4}$$

$$\text{From (4): } 2 \times \frac{105}{4} - 4q_2 - 5 = 0$$

$$q_2^* = \frac{95}{8}$$

$$\text{From (1): } \lambda^* = 4 + 2q_2^* = 28$$

Stationary point is $(\frac{105}{4}, \frac{95}{8}, 28)$

$$L_{11} = 0, L_{12} = 4, L_{21} = 4, L_{22} = 0, q_1 = 2, q_2 = 4$$

$$D = 0 \cdot 4^2 - 2 \cdot 4 \cdot 2 \cdot 4 + 0 \cdot 2^2 = -64 < 0$$

$\Rightarrow (\frac{105}{4}, \frac{95}{8})$ is a local maximum

Profit increases by $\lambda = 28$ when time increases by one unit

Tutorial Q3: Setting up profit and FOCs

Knowledge used

- ▶ Construct the objective function before forming the Lagrangian.
- ▶ Use FOCs for constrained profit maximization.

Solution

First,

$$\pi(q_1, q_2) = (15q_1 + 12q_2) - (20 - 4q_1q_2 + 7q_1 + 6q_2) = 8q_1 + 6q_2 + 4q_1q_2 - 20.$$

The Lagrangian is

$$\mathcal{L} = 8q_1 + 6q_2 + 4q_1q_2 - 20 - \lambda(2q_1 + 4q_2 - 100).$$

The FOCs are

$$8 + 4q_2 - 2\lambda = 0, \quad 6 + 4q_1 - 4\lambda = 0, \quad 2q_1 + 4q_2 = 100.$$

Tutorial Q3: Solving for output

Knowledge used

- ▶ Solve the FOCs by expressing λ from one equation.
- ▶ If integer output is required, choose the best feasible integer bundle near the continuous optimum.

Solution

From the first FOC,

$$\lambda = 4 + 2q_2.$$

Substitute into the second FOC:

$$6 + 4q_1 - 4(4 + 2q_2) = 0 \Rightarrow q_1 = \frac{5}{2} + 2q_2.$$

Substitute into the constraint:

$$2\left(\frac{5}{2} + 2q_2\right) + 4q_2 = 100 \Rightarrow q_2 = \frac{95}{8}, \quad q_1 = \frac{105}{4}.$$

If output must be integer-valued, use the feasible integer bundle

$$q_1^* = 26, \quad q_2^* = 12.$$

Tutorial Q3: Profit, SOC, and shadow price

Knowledge used

- ▶ Substitute the optimal bundle into the objective function.
- ▶ Use $D < 0$ to confirm a local maximum.
- ▶ Interpret λ as the shadow price of the constraint resource.

Solution

Using the integer solution,

$$\pi_{\max} = 8(26) + 6(12) - 20 + 4(26)(12) = 1508.$$

For the SOC,

$$\mathcal{L}_{11} = 0, \quad \mathcal{L}_{22} = 0, \quad \mathcal{L}_{12} = 4, \quad g_1 = 2, \quad g_2 = 4.$$

$$D = 0(4)^2 - 2(4)(2)(4) + 0(2)^2 = -64 < 0,$$

so this is a local maximum.

At the integer solution,

$$\lambda = 4 + 2q_2 = 4 + 2(12) = 28.$$

This means one more unit of the constrained resource, time, increases maximum profit by approximately 28 units.

Question 4

Find the demand functions for x_1 and x_2 from

$$\max_{x_1, x_2} U(x_1, x_2) = 2 \ln(x_1) + \ln(x_2 - 2)$$

5 min

subject to the budget constraint

$$y = p_1 x_1 + p_2 x_2.$$

Then evaluate demand when $p_1 = 4$, $p_2 = 2$, and $y = 100$.

Question 4

Find the demand functions for x_1 and x_2 from

$$\max_{x_1, x_2} U(x_1, x_2) = 2 \ln(x_1) + \ln(x_2 - 2)$$

5 min

subject to the budget constraint

$$y = p_1 x_1 + p_2 x_2.$$

Then evaluate demand when $p_1 = 4$, $p_2 = 2$, and $y = 100$.

Consumer find demand for x_1, x_2

Your Goal: max level of happiness $\Leftrightarrow U(x_1, x_2) = 2 \ln(x_1) + \ln(x_2 - 2)$

given money you have and price of x_1, x_2 . constraint: Income = money you spend on goods $\Leftrightarrow y = p_1 x_1 + p_2 x_2$

$$L = 2 \ln(x_1) + \ln(x_2 - 2) - \lambda (p_1 x_1 + p_2 x_2 - y)$$

Stationary point is $(\frac{2(y-2p_2)}{3p_1}, \frac{y+4p_2}{3p_2}, \frac{3}{y-2p_2})$

$$\frac{\partial L}{\partial x_1} = \frac{2}{x_1} - \lambda p_1 = 0 \Leftrightarrow \lambda = \frac{2}{p_1 x_1} \quad (1)$$

$$L_{11} = -\frac{2}{x_1^2} < 0, L_{12} = 0, L_{21} = 0, L_{22} = \frac{-1}{(x_2-2)^2} < 0$$

$$\frac{\partial L}{\partial x_2} = \frac{1}{x_2 - 2} - \lambda p_2 = 0 \Leftrightarrow \lambda = \frac{1}{p_2(x_2 - 2)} \quad (2)$$

$$D = L_{11} \cdot L_{22} - L_{12}^2 = \frac{2}{x_1^2(x_2-2)^2} > 0$$

$$\frac{\partial L}{\partial \lambda} = -(p_1 x_1 + p_2 x_2 - y) = 0 \quad (3)$$

$\Rightarrow L$ is concave

$\Rightarrow (\frac{2(y-2p_2)}{3p_1}, \frac{y+4p_2}{3p_2}, \frac{3}{y-2p_2})$ is a global maximum

From (1) and (2): $\frac{2}{p_1 x_1} = \frac{1}{p_2(x_2 - 2)}$

$$x_1 = \frac{2p_2(x_2 - 2)}{p_1} \quad (4)$$

plug $p_1 = 4, p_2 = 2, y = 100$ into $\Rightarrow$

$$x_1 = 16$$

$$x_2 = 18$$

plug (4) into (3): $\frac{2p_2(x_2 - 2)}{p_1} \cdot p_1 + p_2 \cdot x_2 = y$

$$2p_2(x_2 - 2) + p_2 x_2 = y$$

$$3p_2 x_2 - 4p_2 = y$$

$$x_2^* = \frac{y + 4p_2}{3p_2}$$

From (4): $x_1^* = \frac{2p_2}{p_1} \cdot \frac{y + 4p_2}{3p_2} - \frac{4p_2}{p_1} = \frac{2(y - 2p_2)}{3p_1}$

From (1): $\lambda^* = \frac{2}{p_1 x_1} = \frac{3}{y - 2p_2}$

function $\Rightarrow$ integer

Tutorial Q4: Lagrangian and FOCs

Knowledge used

- ▶ Use the Lagrange method to derive demand as functions of income and prices.
- ▶ Remember the domain: $x_1 > 0$ and $x_2 > 2$.

Solution

$$\mathcal{L}(x_1, x_2, \lambda) = 2 \ln(x_1) + \ln(x_2 - 2) - \lambda(p_1 x_1 + p_2 x_2 - y).$$

The FOCs are

$$\frac{2}{x_1} - \lambda p_1 = 0, \quad \frac{1}{x_2 - 2} - \lambda p_2 = 0, \quad p_1 x_1 + p_2 x_2 - y = 0.$$

From the budget constraint,

$$x_1 = \frac{y - p_2 x_2}{p_1}.$$

From the first FOC,

$$\lambda = \frac{2}{p_1 x_1}.$$

Tutorial Q4: Demand functions

Knowledge used

- ▶ Substitute the budget expression into the FOC ratio.
- ▶ Confirm the stationary point using the local SOC.

Solution

Substituting into the second FOC gives

$$\frac{1}{x_2 - 2} = \frac{2p_2}{y - p_2x_2}.$$

Hence

$$y - p_2x_2 = 2p_2(x_2 - 2) \Rightarrow y + 4p_2 = 3p_2x_2.$$

Therefore,

$$x_2^* = \frac{y + 4p_2}{3p_2}, \quad x_1^* = \frac{2(y - 2p_2)}{3p_1}.$$

The local SOC gives

$$D = -2 \left(\frac{p_2}{x_1} \right)^2 - \left(\frac{p_1}{x_2 - 2} \right)^2 < 0,$$

so the stationary point is a local maximum.

Tutorial Q4: Numerical demand

Knowledge used

- Substitute prices and income into the demand functions.

Solution

When $p_1 = 4$, $p_2 = 2$, and $y = 100$,

$$x_1^* = \frac{2(100 - 2 \cdot 2)}{3 \cdot 4} = \frac{192}{12} = 16,$$

$$x_2^* = \frac{100 + 4 \cdot 2}{3 \cdot 2} = \frac{108}{6} = 18.$$

Therefore,

$$(x_1^*, x_2^*) = (16, 18).$$

Key takeaways

- ▶ A constrained optimization problem combines an objective function with a constraint.
- ▶ The Lagrangian method turns the constrained problem into a system of FOCs.
- ▶ The multiplier λ is a shadow price: it measures the marginal value of relaxing the constraint.
- ▶ Use local SOC's for classification and concavity/convexity for global conclusions.